

4 Front-End Development: Structure & Style

- HTML5: Student should go through the material on HTML5
- CSS3 (Cascading Style Sheets)
- Framework: Introduction to Bootstrap for responsive front-end design

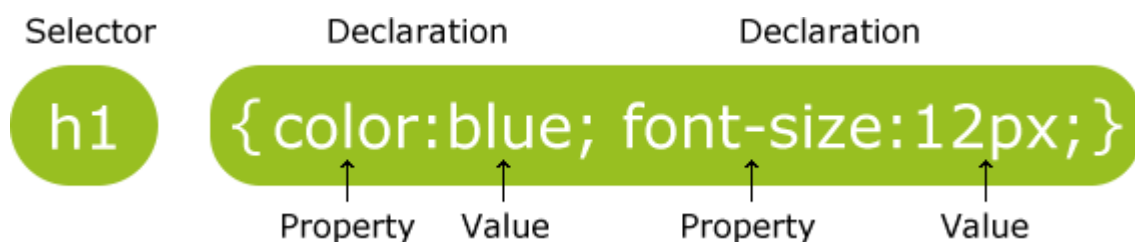
4.1 Cascading Style Sheets (CSS)

4.1.1 Intro to CSS

- HTML was designed to specify the semantics (i.e., content and structure) of a document
 - e.g., which parts are elements of section headers, body text, links, lists, tables, etc.)
 - But HTML markup can also be used to indicate not only semantics of a document, but also some presentation (fonts, spacing, sizes, colours, positioning, etc.)
 - It is better though, not to mix presentation with content
 - Use markup predominantly for indicating the semantics of a document and use a separate mechanism to determine exactly how information contained in the document should be presented
- CSS provide a mechanism to separate semantics from presentation
- This separation of structure from presentation simplifies maintaining and modifying web pages, especially on large-scale websites
 - If a website's presentation is determined entirely by a style sheet, you can simply swap in a new style sheet to completely change the site's appearance

4.1.2 CSS Rule (Syntax)

- Comprises
 - A selector
 - The HTML element you want to style (or format)
- A declaration block
 - One or more declarations separated by semicolons
 - Each declaration includes a property name and a value, separated by a colon
 - Declaration always ends with a semicolon, and declaration groups are surrounded by curly braces



- CSS comment
 - Starts with `/*` and ends with `*/`
- CSS Selectors
 - Used to select the elements to be styled
 - Select elements based on their names, id, class

4.1.3 Declaring Document Styles

- Three approaches
 - Inline styles
 - Embedded style sheets
 - External style sheets

4.1.3.1 Inline Styles

- Declare an individual element's format using the HTML5 **style** attribute
- Override any other styles applied using embedded styles sheets and external style sheets
- Note: Inline styles do not truly separate presentation from content

```
<!DOCTYPE html>
<html>
<head>
<title>Inline Styles</title>
</head>
<body>
<p>This text does not have any style applied to it.</p>
<p style = "font-size: 20pt;">This text has the <em>font-size</em> style
applied to it, making it 20pt.</p>
<p style = "font-size: 20pt; color: deepskyblue;">This text has the <em>font-
size</em> and <em>color</em> styles applied to it, making it
20pt and deep sky blue.</p>
</body>
</html>
```

This text does not have any style applied to it.

This text has the *font-size* style applied to it, making it 20pt.

This text has the *font-size* and *color* styles applied to it, making it 20pt and deep sky blue.

4.1.3.2 Embedded style sheets

- Embed a CSS3 document in an HTML document's head section
- Styles are placed in the head element and apply to matching elements wherever they appear in the body

```
<!DOCTYPE html>
<html>
<head>
  <title>Embedded Style Sheet</title>
  <style type = "text/css">
    em { font-weight: bold;
          color: black; }
    h1 { font-family: tahoma, helvetica, sans-serif; }
    p { font-size: 12pt;
        font-family: arial, sans-serif; }
    .special { color: purple; }
  </style>
</head>
<body>
  <h1 class = "special">Deitel & Associates, Inc.</h1>
  <p>Deitel & Associates, Inc. is an authoring and corporate training
organization specializing in programming languages, Internet and web
technology, iPhone and Android app development, and object technology
education.</p>
  <h1>Clients</h1>
  <p class = "special"> The company's clients include many <em>Fortune 1000
companies</em>, government agencies, branches of the military and business
organizations.</p>
</body>
</html>
```

Deitel & Associates, Inc.

Deitel & Associates, Inc. is an authoring and corporate training organization specializing in programming languages, Internet and web technology, iPhone and Android app development, and object technology education.

Clients

The company's clients include many **Fortune 1000 companies**, government agencies, branches of the military and business organizations.

- Some notes
 - **font-family** Property
 - Not all users have the same fonts installed on their computers

Generic font families	Examples
serif	times new roman, georgia
sans-serif	arial, verdana, futura
cursive	script
fantasy	critter
monospace	courier, fixedsys

- You to specify a comma-separated list of fonts to use for a particular style
 - Browser attempts to use the fonts in the order in which they appear in the list
 - Advisable to end a font list with a generic font family name in case the other fonts are not installed on the user's computer

4.1.3.3 *External style sheets*

- Separate document that contain only CSS rules
- Enables us provide a uniform look and feel to an entire website (or to a portion of one)
- Can reuse the same external style sheet across multiple websites
- Different pages on a site can all use the same style sheet
- When changes to the styles are required, you need to modify only a single CSS file to make style changes across all the pages that use those styles

```

styles.css

body { font-family: arial, helvetica, sans-serif; }
a.nodect { text-decoration: none; }
a:hover { text-decoration: underline; }
li em { font-weight: bold; }
h1, em { text-decoration: underline; }
ul { margin-left: 20px; }
ul ul { font-size: .8em; }

```

```

<!DOCTYPE html>
<html>
  <head>
    <title>Linking External Style Sheets</title>
    <link rel = "stylesheet" type = "text/css" href = "styles.css">
  </head>
  <body>
    <h1>Shopping list for <em>Monday</em>:</h1>
    <ul>
      <li>Milk</li>
      <li>Bread
        <ul>
          <li>white bread</li>
          <li>Rye bread</li>
          <li>Whole wheat bread</li>
        </ul>
      </li>
      <li>Carrots</li>
      <li>Yogurt</li>
      <li>Pizza <em>with mushrooms</em></li>
    </ul>
    <p><em>Go to the</em>
      <a class = "nodect" href = "http://www.deitel.com">Grocery store</a>
    </p>
  </body>
</html>

```

Shopping list for *Monday*:

- Milk
- Bread
 - white bread
 - Rye bread
 - Whole wheat bread
- Carrots
- Yogurt
- *Pizza with mushrooms*

Go to the [Grocery store](#)

Some notes on the styling

- | | |
|-------------|--|
| 1. body | { font-family: arial, helvetica, sans-serif; } |
| 2. a.nodect | { text-decoration: none; } |
| 3. a:hover | { text-decoration: underline; } |
| 4. li em | { font-weight: bold; } |
| 5. h1, em | { text-decoration: underline; } |
| 6. ul | { margin-left: 20px; } |
| 7. ul ul | { font-size: .8em; } |

Line 2

- Applies property text-decoration to all **a** elements whose class attribute is set to **nodect**. Consequently, the browser will not underline the hyperlink
- Other possible values for text-decoration: overline, line-through, and underline. The .nodect appended to a is a more specific class selector; this style in line

Line 3

- Specifies a style for hover, which is a **pseudo-class**
 - Pseudo-classes give you access to information that is not declared in the document, such as whether the mouse is hovering over an element or whether the user has previously clicked (visited) a particular hyperlink
- The hover pseudo-class is activated dynamically when the user moves the mouse cursor over (i.e., hovers over) an element
- Pseudo-classes are separated by a colon (with no surrounding spaces) from the name of the element to which they are applied

Line 4

- Causes all **em** elements that are children of **li** elements to be bold
 - em elements that are not nested within li elements will be formatted using the style for these em elements

Line 5

- Selectors separated with a comma to apply an underline style rule to all h1 and all em elements

Line 6: Assigns a 20-pixel left margin to all ul elements

Line 7: Applies the font size to all unordered lists nested within an unordered list.

4.1.4 CSS Tutorials and Readings

- <https://www.w3schools.com/css/default.asp>
- Web Technologies: A Computer Science Perspective
 - Chapter 3: Style Sheets - CSS
- Internet & World Wide Web: How to Program
 - Introduction to Cascading Style Sheets (CSS): Part 1
 - Introduction to Cascading Style Sheets (CSS): Part 2 (assorted examples, including, Fig 4.14 (borders), Fig 4.15 (box model and text flow), Fig 4.16 (media types), Fig 4.17 (Drop Down Menus))
 - If time permits, introduce CSS Flexbox and CSS Grid, CSS Responsive

4.2 Introduction to Bootstrap for responsive front-end design

See https://www.w3schools.com/bootstrap5/bootstrap_get_started.php

What is Bootstrap?

- An open-source front-end framework used to build responsive, mobile-first web applications
- Provides a collection of pre-designed HTML and CSS templates for interface elements like typography, forms, buttons, and navigation, along with optional JavaScript plugins for interactive features.
- Core Features
 - Responsive Grid System: Uses a 12-column flexbox-based grid that automatically reorders or resizes elements to fit desktops, tablets, and smartphones
 - Pre-built Components: Includes ready-to-use UI elements such as
 - Navigation: Navbars, breadcrumbs, and pagination
 - Forms: Styled input fields, checkboxes, and validation features
 - Interactivity: Modals, carousels, tooltips, and dropdowns powered by JavaScript
 - Mobile-First Approach: Styles are optimized for mobile devices by default and scaled up for larger screens using CSS media queries

Getting Bootstrap

- Either download from getbootstrap.com or include it from a CDN (Content Delivery Network) like jsDelivr
- Including from a CDN is most common usage. Why?
 - Many users already have downloaded Bootstrap 5 from jsDelivr when visiting another site

- Hence, Bootstrap 5 will be loaded from cache when users visit your site, which leads to faster loading time
- Also, most CDNs will make sure that once a user requests a file from it, it will be served from the server closest to them, which also leads to faster loading time
- How to include Bootstrap 5?

```
<!-- Latest compiled and minified CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Latest compiled JavaScript -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"></script>
```

Key steps in using Bootstrap

- Add the HTML5 doctype

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Bootstrap 5 Example</title>
    <meta charset="utf-8">
  </head>
</html>
```

- Mobile-first
 - To ensure proper rendering and touch zooming, add the following <meta> tag inside the <head> element

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

- width=device-width: Sets the width of the page to follow the screen-width of the device (which will vary depending on the device)
- initial-scale=1: sets the initial zoom level when the page is first loaded by the browser
- Containers
 - A containing element is used to wrap site contents
 - 2 container classes to choose from
 - .container class provides a responsive fixed width container
 - .container-fluid class provides a full width container, spanning the entire width of the viewport

Bootstrap examples

- Bootstrap 5 Grid System
 - Built with flexbox and allows up to 12 columns across the page
 - If you do not want to use all 12 columns individually, you can group the columns together to create wider columns

span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1	span 1
span 4				span 4				span 4			
span 4				span 8							
span 6						span 6					
span 12											

- The grid system is responsive, and the columns will re-arrange automatically depending on the screen size
- Example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
rel="stylesheet">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"><
  /script>
</head>
<body>

<div class="container-fluid mt-3">
  <h1>Three equal width columns</h1>
  <p>Note: Try to add a new div with class="col" inside the row class - this will
create four equal-width columns.</p>
  <div class="row">
    <div class="col p-3 bg-primary text-white">.col</div>
    <div class="col p-3 bg-dark text-white">.col</div>
    <div class="col p-3 bg-primary text-white">.col</div>
  </div>
</div>
```

- Note: mt-3 top margin of 3; p-3 padding of 3

Three equal width columns

Note: Try to add a new div with class="col" inside the row class - this will create four equal-width columns.




```

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href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
rel="stylesheet">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"><
/script>
</head>
<body>

<div class="container-fluid mt-3">
  <h1>Responsive Columns</h1>
  <p>Resize the browser window to see the effect.</p>
  <p>The columns will automatically stack on top of each other when the screen is
less than 576px wide.</p>
  <div class="row">
    <div class="col-sm-3 p-3 bg-primary text-white">.col</div>
    <div class="col-sm-3 p-3 bg-dark text-white">.col</div>
    <div class="col-sm-3 p-3 bg-primary text-white">.col</div>
    <div class="col-sm-3 p-3 bg-dark text-white">.col</div>
  </div>
</div>

</body>
</html>

```

- On mobile phones or screens that are less than 576px ("col-sm-3) wide, the columns will automatically stack on top of each other

Responsive Columns

Resize the browser window to see the effect.

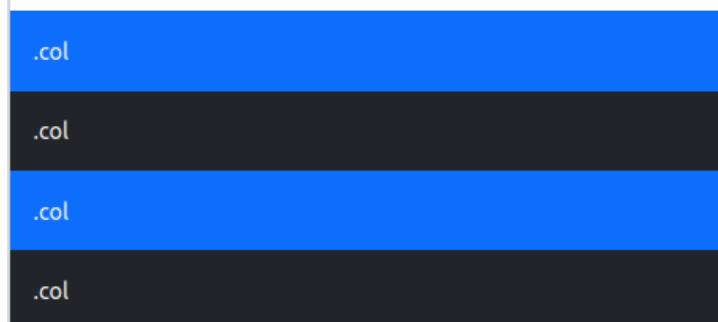
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Responsive Columns

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```

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  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
rel="stylesheet">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"><
/script>
</head>
<body>

<div class="container-fluid mt-3">
  <h1>Two Unequal Responsive Columns</h1>
  <p>Resize the browser window to see the effect.</p>
  <p>The columns will automatically stack on top of each other when the screen is
less than 576px wide.</p>
  <div class="row">
    <div class="col-sm-4 p-3 bg-primary text-white">.col</div>
    <div class="col-sm-8 p-3 bg-dark text-white">.col</div>
  </div>
</div>

</body>
</html>

```

Two Unequal Responsive Columns

Resize the browser window to see the effect.

The columns will automatically stack on top of each other when the screen is less than 576px wide.

